

CarCodex Full Project Handbook v1.0

Executive Summary

CarCodex is a local-first vehicle ownership, maintenance and document archival platform. Primary goal: data remains understandable even if the application disappears.

Architecture

UI Layer -> Routers -> Database -> Storage Layer -> Filesystem. UI templates render workflows. Routers coordinate business logic. SQLite stores metadata. storage.py is the authoritative filesystem abstraction.

Core Folders

CarCodex_Data/Vehicles/___V/ containing VehicleInfo.txt, Documents, Maintenance, Photos and Exports.

Database Models

Vehicle, MaintenanceVisit, ServiceRecord, Document, Vendor, ServiceType, MaintenanceSchedule, MileageHistory, Event, Snapshot, OwnershipCost.

Storage Layer

safe_name, get_vehicle_folder, create_vehicle_folders, create_vehicle_info_file, get_document_folder, build_document_filename, build_document_path, save_document_file, get_maintenance_folder.

Current Workflows

Vehicle creation automatically creates storage folders and VehicleInfo.txt. Maintenance creation automatically creates MaintenanceVisit and ServiceRecord. Invoice upload stores documents inside maintenance-event folders.

Router Responsibilities

vehicle_ui.py owns user workflows. uploads.py handles document ingestion. documents.py manages document records. maintenance-related routers manage maintenance functions.

AI Onboarding

Always treat helpers/storage.py as the single source of truth for filesystem decisions. Never hardcode paths. Create records first when IDs are required for filenames. Preserve human-readable storage. Favor workflow reduction. Minimize user navigation. Keep local-first design. Vehicle folders must remain stable using Year_Make_Model_VehicleID. VehicleInfo.txt is authoritative metadata on disk.

Roadmap v1.1

Automatically create Document records from maintenance invoices. Maintenance detail page. Document detail page. Maintenance/Documents integration.

Roadmap v1.5

Downloads, dashboard improvements, backup/export system, UI cleanup and workflow refinement.

Roadmap v2.0

OCR, AI summaries, reporting, recommendations, advanced document classification.

Filesystem Structure

```
CarCodex_Data/  
■ Vehicles/  
■ 2008_Chevrolet_Corvette_V17/  
■ VehicleInfo.txt  
■ Documents/  
■ Maintenance/  
■ ■ 2026-08-02-Oil_Change-M52/  
■ ■ Invoice.pdf  
■ Photos/  
■ Exports/
```